FORCE & JACOBIANS FOR A SPRING BETWEEN 2 NODES

CHRONO - FORMULATION

$$F = k(\ell - \ell_0)\, d + r\, \dot{\ell}\, d$$

Let $v = p_2 - p_1$

$\dot{v} = \dot{p}_2 - \dot{p}_1$

$\ell^2 = v^T v$, $\dot{\ell} = \dot{v}^T d$

$d = v/\ell \Leftrightarrow \ell d = v$

Want derivatives w.r.t.

$\alpha = p_1$ or p_2

$\beta = \dot{p}_1$ or $\dot{p}_2$

for which

$$\frac{\partial v}{\partial \alpha} = \begin{cases} -I, & \text{if } \alpha = p_1 \\ +I, & \text{if } \alpha = p_2 \end{cases}$$

$$\frac{\partial \dot{v}}{\partial \beta} = \begin{cases} -I, & \text{if } \beta = \dot{p}_1 \\ +I, & \text{if } \beta = \dot{p}_2 \end{cases}$$

- $2\ell \frac{\partial \ell}{\partial \alpha} = 2 v^T \frac{\partial v}{\partial \alpha} \Rightarrow \boxed{\frac{\partial \ell}{\partial \alpha} = d^T \frac{\partial v}{\partial \alpha}}$

- $\ell \frac{\partial d}{\partial \alpha} + d \frac{\partial \ell}{\partial \alpha} = \frac{\partial v}{\partial \alpha}$

$$\frac{\partial d}{\partial \alpha} = \frac{1}{\ell}\left[\frac{\partial v}{\partial \alpha} - d d^T \frac{\partial v}{\partial \alpha}\right] \Rightarrow \boxed{\frac{\partial d}{\partial \alpha} = \frac{1}{\ell}\left[\frac{\partial v}{\partial \alpha} - D\right]\frac{\partial v}{\partial \alpha}} \qquad D \triangleq d d^T$$

- $\dot{\ell} = \dot{v}^T d \Rightarrow \boxed{\frac{\partial \dot{\ell}}{\partial \beta} = d^T \frac{\partial \dot{v}}{\partial \beta}}$

- $\ell d = v \Rightarrow \dot{\ell} d + \ell \dot{d} = \dot{v} \Rightarrow \dot{d} = (\dot{v} - \dot{\ell} d)/\ell$

- $\ell \dot{\ell} = \dot{v}^T v \Rightarrow \frac{\partial \ell}{\partial \alpha}\dot{\ell} + \ell \frac{\partial \dot{\ell}}{\partial \alpha} = \dot{v}^T \frac{\partial v}{\partial \alpha} \Rightarrow \frac{\partial \dot{\ell}}{\partial \alpha} = \frac{1}{\ell}\left[\dot{v}^T \frac{\partial v}{\partial \alpha} - \dot{\ell} d^T \frac{\partial v}{\partial \alpha}\right]$

$$\Rightarrow \boxed{\frac{\partial \dot{\ell}}{\partial \alpha} = \dot{d}^T \frac{\partial v}{\partial \alpha}} \qquad \text{Define } \dot{D} \triangleq d \dot{d}^T$$

$$\boxed{\frac{\partial F}{\partial \alpha} = \left[k D + \frac{k(\ell - \ell_0) + r\dot{\ell}}{\ell}(I - D) + r\dot{D}\right]\frac{\partial v}{\partial \alpha}}$$

$$\boxed{\frac{\partial F}{\partial \beta} = r D \frac{\partial \dot{v}}{\partial \beta}}$$

$$\hat{J}_{\alpha\beta} = K_{factor}\frac{\partial F}{\partial \alpha} + R_{factor}\frac{\partial F}{\partial \beta}$$